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T - Test

A sample of 10 students scored the following marks in an exam: [72, 68, 75, 70, 74, 69, 71, 73, 70, 72]
We want to test whether the average mark = 70 ($\mu_0 = 70$) at 5% significance level using python

```
import numpy as np from
scipy import stats
marks = np.array([72, 68, 75, 70, 74, 69, 71, 73, 70, 72])
mu_0 = 70
t_stat, p_value = stats.ttest_1samp(marks, mu_0) print(f"T-
statistic: {t_stat:.3f}") print(f"P-value: {p_value:.4f}") alpha =
0.05 if p_value < alpha: print("Reject Null Hypothesis → Mean
is significantly different from 70.") else: print("Fail to
Reject Null Hypothesis → No significant difference.")
T-statistic: 1.993
P-value: 0.0774

Fail to Reject Null Hypothesis → No significant difference.
```

Z - Test

```
Z-statistic: 2.400
P-value: 0.0164

Reject Null Hypothesis → Mean is significantly different from 50 g.
```

Anova Test

```
import numpy as np from math
import sqrt from scipy.stats
import norm x_bar = 51.2
mu_0 = 50 sigma = 3 n = 36
z_stat = (x_bar - mu_0) / (sigma / sqrt(n)) p_value = 2 * (1 -
norm.cdf(abs(z_stat))) print(f"Z-statistic: {z_stat:.3f}")
print(f"P-value: {p_value:.4f}") alpha = 0.05 if p_value < alpha:
print("Reject Null Hypothesis → Mean is significantly different
from 50 g.") else: print("Fail to Reject Null Hypothesis → No
significant difference.")
```

Three fertilizers (A, B, C) were tested on crop yield (in kg). Is there a significant difference among fertilizers? (Use $\alpha = 0.05$)

Fertilizer Yields

A 20, 22, 23

B 19, 20, 18

C 25, 27, 26

```
import numpy as np from
scipy import stats
A=[20,22,23]
B=[19,20,18]
C=[25,27,26]
f_stat,p_value=stats.f_oneway(A,B,C) print(f"F-statistic:
{f_stat:.3f}") print(f"P-value: {p_value:.4f}") alpha=0.05 if
p_value<alpha: print("Reject Null Hypothesis-Means are
significantly different.") else: print("Fail to Reject Null Hypo
```